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DIAGNOSTIC FIELD GUIDE

Blower Won't Hit Airflow Setpoint: A Field Guide for Composting and Digester Aeration

Field notes from Jonathan Gilmour, independent controls engineer, Northern California.

You've got an aerated static pile (ASP) or in-vessel bay calling for a set number of standard cubic feet per minute (SCFM), the blower is running, and the airflow display just sits there below command. Or a mixing/recirculation blower on a digester won't reach its flow target and the process operator is watching temperature or mixing drift the wrong way. Before anyone starts tuning the PID (proportional-integral-derivative controller) or swapping a transmitter, this guide walks the causes in the order I actually check them at a plant: the demand side first, because a blower fighting a plugged pile or a closed damper is the most common call I get, and it is almost never a controls problem. Numbers here are typical ranges to orient you. Check every one against your machine's nameplate, your blower curve, and your control narrative before you act.

A note that matters before anything else: on composting and digester sites you are working around biology that makes gas. Read the safety block below before you open a plenum, a gas holder, or a vault.

Gas safety and confined space — read before you go hands-on

Compost piles, digesters, gas holders, condensate sumps, and pump/blower vaults on these sites are permit-required confined spaces and often a biogas atmosphere. Do not treat this as boilerplate.

- **Hydrogen sulfide (H2S) kills fast.** It deadens your sense of smell around 100 ppm, which is also roughly the IDLH (Immediately Dangerous to Life or Health). If you can smell rotten egg and then can't, that is not "it cleared," that is your nose failing. Digester headspace, condensate legs, and stagnant plenums under a wet pile can pocket H2S.
- **Methane is explosive** from about 5% (lower explosive limit, LEL) to about 15% (upper explosive limit, UEL) by volume in air. A leaking gas holder, a cracked pile emitting off a digester feed, or a blower pulling from an enclosed source can put you in that band.

- **Never introduce air into an anaerobic (biogas) digester.** On an anaerobic digester the "airflow setpoint" is on a gas mixing/recirculation blower moving recirculated biogas, not air — pushing air into that headspace drives the vessel atmosphere through the 5–15% methane band inside the tank, where you can't ventilate it out. The air-aeration content in this guide belongs to aerobic systems (ASP composting, aerobic sludge digestion). A biogas-mixing blower carries the flammable atmosphere in the machine itself, so it sits in a hazardous-area classification and every point below about worn wheels and open enclosures applies with that in mind.
- Wear a **personal 4-gas monitor** (H_2S, LEL, O_2, CO), bump-tested that shift, on your body — not set on a rail ten feet away.
- **Atmospheric-test and get the permit** before entry into any vault, plenum, digester, or gas holder. Test top, middle, bottom; gases stratify.
- Biogas areas are typically **Class I, Division 1 or 2** (or Zone 0/1/2) per NEC 500/505. Your meter, light, and tools in those areas must match the area classification. A standard laptop or phone is an ignition source in a Div 1 space.

Lockout/tagout — before any hands-on work on the blower

Aeration blowers are rotating equipment, and many restart on their own from the PLC (programmable logic controller) or on a timer. Before you touch a coupling, a belt, an impeller inlet, or a damper linkage:

- Establish a **zero-energy state**: disconnect and lock the blower motor at the local disconnect, and lock the VFD (variable frequency drive) so it cannot command a start. Tag it.
- **Verify dead** — try-to-start at the HMI (human-machine interface) and confirm no rotation, then confirm zero voltage at the motor terminals with a meter you just proved on a known source.
- Blowers coast a long time and can **windmill backward** on system draft. Confirm the shaft is stopped and blocked before your hand goes near the inlet.
- Where a step below involves an energized test (checking VFD output, transmitter signal at a live terminal), that is a live-electrical task — proper PPE (personal protective equipment), one hand, and a second person aware. The hazard warning belongs above that work, so: **any measurement taken inside an energized enclosure is an arc-flash exposure. Dress for it or don't open the door.**

10-minute triage

Fast narrowing before you commit tools. You're trying to bin this into demand / machine / measurement / control.

1. **Read three numbers at the HMI:** airflow feedback (SCFM), the pressure feedback (system static, inches water column "in. w.c." or the digester's discharge pressure in psi), and VFD speed/load

(Hz and % current or torque). Write them down with the time.

2. **Is the VFD pinned at max Hz** (or at a current/torque limit) while airflow stays low? That points machine or demand side, not controls. Note whether it's at 60 Hz *and* riding a current limit (overloaded) versus at 60 Hz with normal current (can't move air — leak or worn machine). How current reads against backpressure depends on the blower type: a positive-displacement (PD) rotary-lobe blower pulls current in step with discharge pressure, so a current limit there usually is real backpressure; a centrifugal doesn't rise cleanly with backpressure (a backward-inclined wheel can draw *less* near shutoff), so confirm with the pressure reading in steps 4–5, not current alone.
3. **Is the VFD *not* at max**, output held down while airflow is under setpoint? That's controls — PID output-limited, in manual, or an interlock capping speed. Go to the control section.
4. **Pressure high, airflow low?** Backpressure. The pile/biofilter/plenum is restricting. Demand side.
5. **Pressure low *and* airflow low, blower at speed?** The machine is spinning but not making pressure — leak, open bypass, worn impeller/clearances, belt slip, or wrong rotation. Machine side.
6. **Feedback frozen, pegged, or impossibly steady** against a changing command? Suspect the transmitter or its tubing before anything else. Measurement side.
7. **Cross-check one number by hand.** A pitot or vane anemometer at a test port, or a second gauge on the header, tells you in two minutes whether the airflow reading is even real. Half the "won't hit setpoint" calls are a lying transmitter.

If 4 sends you to demand and 5 sends you to machine, trust the pressure signal — it separates "can't push against the load" from "isn't pushing at all."

Symptom → likely cause → check (and log)

Symptom	Likely cause	Check — and what to log
Airflow low, **static pressure climbing** over days/weeks	Biofilter or media differential pressure rising; wet/matted media; fines blinding the bed	Log the media dP (differential pressure) trend. Compare today's dP at a given SCFM to commissioning baseline. Rising dP at equal flow = load side, not controls.
Airflow low, pressure high, **one zone worse than others**	Plenum plugged, condensate flooding a leg, or pile short-circuiting (channeling)	Walk the plenum drains. Probe pile for hot/cold and wet spots. Log which zone, drain volume pulled.
Damper **commanded open, airflow says closed**	Actuator not tracking command; seized linkage; failed feedback pot	Compare damper command % to actual blade position by eye. Log command vs. observed. Manually stroke if safe.

Symptom	Likely cause	Check — and what to log
VFD at **max Hz, riding current/torque limit**	Blower overloaded — high backpressure, or motor/drive derated	Log Hz, % current, % torque, drive temp. Current-limited = load too high or drive fault, not a setpoint you can dial in.
VFD at **max Hz, normal current, low pressure**	Worn impeller/clearances, belt slip, open bypass/relief, or a big air leak	Feel/listen for slip. Check bypass/relief valve seated. Log discharge pressure vs. blower curve at that Hz.
Blower runs, **near-zero pressure and flow**	Wrong rotation (rewired after work), or backward-installed impeller	Confirm rotation arrow. Log phase rotation if recently serviced.
Airflow feedback **frozen or drifting** while command changes	Transmitter fouled, condensate-blocked sensing line, drifted zero, wrong scaling	Blow down/verify sensing lines. Compare to handheld. Log raw mA and engineering-unit reading.
Airflow **oscillating / hunting** around setpoint	PID too aggressive for a slow process; damper and VFD loops fighting	Log the swing period and amplitude. Slow the loop before you touch gains twice.
Airflow parks **just under setpoint, output maxed**	PID output-limited or wound up; setpoint above what the system can deliver	Log PID output %. If output is at 100% and flow still short, the loop is done its job — problem is downstream.
Temperature and oxygen **loops disagree**, airflow chasing both	Temp-vs-O2 control fighting; cascade misconfigured	Log both setpoints and both feedbacks. Determine which loop is master right now.

Demand / backpressure side — check this first

Most "won't hit setpoint" calls on ASP and in-vessel systems are the load, not the blower or the code.

Media / biofilter differential pressure climbing. ASP odor control biofilters and the compost media itself load up over time. Wet media, compaction, and fines migration all raise the pressure the blower must push against. Pull up the dP trend across the bed. If today's dP at a given SCFM is well above the commissioning baseline, the blower is doing what it can against a bed that's tightening up. That's a media job — remoisten to spec and rip/refluff, or rebuild the bed — not a gain change. Log the dP-vs-flow point so you can prove the trend to whoever owns the media.

Wet or matted media, condensate in the plenum. Cold nights and a hot pile drop condensate in the ductwork and plenum. A flooded plenum leg or a low spot full of water is a hidden restriction that reads as "blower won't make flow." Walk the drains. If you pull a bucket out of a leg that should be dry, you found it.

Plugged plenum or pile short-circuiting. Air takes the easy path. If a pile has channeled — cracked, settled unevenly, or has a wet cap — air blows out the channels and the bed reads low flow at the sensor even while the blower moves plenty of air. Probe the pile for hot and cold zones and wet spots. Uneven temperature across a bay that shares one blower is the tell.

Damper/actuator position vs. command; seized linkage. A modulating damper that's commanded 80% but sitting at 20% will starve the header no matter what the blower does. Stand at the damper and compare the command on the HMI to the actual blade angle. A seized linkage, a slipped clamp on the shaft, a failed feedback potentiometer, or a spring-return actuator that lost air/power will all lie to the controller. This is a five-minute look that saves an hour of loop tuning.

Machine side — the blower itself

You've cleared the load. Now separate "spinning but not pushing" from "can't push against the load."

VFD limiting on current or torque. If the drive is at max Hz and riding a current or torque limit, it is protecting the motor — you cannot dial in more flow with a setpoint. Either backpressure is too high (go back to demand side) or the drive/motor is derated (hot drive, failing motor, phase imbalance). Read the drive's active-limit status word and its fault/warning log. Log Hz, current %, torque %, and drive temperature. Mind the blower type when you read that current: on a positive-displacement (PD) rotary-lobe blower the motor current tracks discharge pressure almost directly, so a current limit at max speed usually is genuine backpressure; on a centrifugal, current does not climb cleanly with backpressure, so lean on the pressure reading, not the amps, to separate load from machine.

Blower runs but makes no pressure = worn impeller/clearances or belt slip. On a PD rotary-lobe blower — the common choice on digester and aeration service — the wear shows as slip past the lobes: it turns at full speed but the differential pressure falls off. On a centrifugal or regenerative blower, worn impeller tips or opened-up running clearances do the same — it spins at full speed and moves little air against the system. On belt-driven units, a glazed or loose belt slips under load — the motor holds speed, the blower shaft doesn't. With the blower safely locked out, check belt tension and look for glazing and dust. Running, compare discharge pressure at a known Hz to the blower curve; well below curve at rated speed points to a worn machine. This is mechanical — a rebuild or belt job, not controls.

Motor / rotation / phase issues. If anyone has been in the motor termination box or the disconnect recently, verify rotation against the arrow. A blower run backward makes a fraction of its flow and pressure and sounds almost right, which fools people. A lost phase or high phase imbalance shows as low output and a hot, growling motor. Confirm rotation and phase before you chase anything subtle.

Measurement side — is the number even real?

Half of these calls are a transmitter, not a process.

Transmitter drifted, fouled, or condensate-blocked. Airflow on these systems is often inferred from a differential-pressure element (orifice, pitot array, averaging tube) or a thermal mass-flow meter. The sensing lines love to fill with condensate and dust in humid compost air, which pinches the reading low or freezes it. Blow down or clear the sensing lines to a drain (isolate first), and confirm the transmitter zeroes with the blower off. A thermal flow element caked with dust reads low. Log the raw signal (mA) alongside the engineering-unit value.

Wrong scaling. After a transmitter swap or a PLC change, the range in the PLC may not match the transmitter's range. A 4–20 mA scaled 0–1000 SCFM in the field but 0–1500 in the code will never agree with reality. Verify the scaling end to end.

Sensor location. A flow or pressure tap too close to an elbow, a damper, or the blower discharge reads turbulence, not process. If the number has always been suspect and the port sits right after a fitting, that's a design issue to note, not a daily fault.

The fastest measurement check is the handheld cross-check from the triage: a vane anemometer or pitot at a test port. If the header truly has flow and the HMI says it doesn't, stop looking at the process and fix the instrument.

Control side — only after the above

Reach for the PID last. On these systems it's usually innocent.

PID wound up or output-limited. If the loop output sits at 100% and airflow is still short, the controller has done everything it can — the restriction is downstream (demand or machine), and no gain change helps. Integral windup shows as a loop that overshoots hard after the restriction finally clears. Check for anti-windup/output clamping in the block. Log the output %.

Airflow setpoint vs. what the pile actually needs. Someone may have set a SCFM target the system was never sized to deliver against current backpressure, or the recipe's airflow is higher than the biology needs. What the pile actually needs is oxygen delivery and heat removal; the SCFM number on the screen is only a proxy for those. If the target is above the fan curve at achievable pressure, the fix is a realistic setpoint or a media/mechanical repair, and no amount of tuning gets there.

Temperature-vs-oxygen loops fighting. Many composting controls run temperature feedback and oxygen feedback, sometimes cascaded, sometimes as competing overrides. If the temp loop wants more air to cool the pile while an O2 high-limit or a runtime cap is pulling air back, airflow chases a moving target and never settles. Find which loop is actually master at this moment, and confirm the override logic isn't sawing against itself. Log both setpoints and both feedbacks together.

The thermal-lag reality — don't over-tune a slow process

Compost and digestion are slow thermal and biological processes. Pile temperature responds to an airflow change over tens of minutes to hours, not seconds. If you tune the airflow loop tight enough to chase every wiggle, you'll fight condensate noise and sensor jitter and never help the biology. Tune the airflow loop to be stable and a little lazy, and let the temperature/oxygen supervisory logic move slowly on top of it. A loop that hunts every few seconds is over-tuned for what the pile can feel.

Timer-aeration fallback. When the feedback is untrustworthy at 2am and the pile still needs air, most well-built ASP systems have a timer-based aeration mode — fixed on/off cycles at a known damper/speed. Drop to timer aeration to keep the biology alive and safe while you sort the instrument or the media in daylight. It is a legitimate, safe fallback, not a failure. Document that you switched to it and why.

When it's mechanical or media, not controls

Call it plainly, because it saves the plant money and the controls tech a bad night: if the static pressure is climbing at equal flow, the answer is **media** — remoisten, refluff, or rebuild. If the blower is at full speed with normal current and still below its curve, the answer is **mechanical** — impeller, clearances, belt, or rotation. If a damper isn't going where it's commanded, the answer is the **actuator or linkage**. None of those get fixed in the PID. The controls are only the culprit when the drive is being held below what the machine and the load can clearly do, or when a loop is genuinely hunting or wound up.

On protective setpoints

If any of the limits capping this loop are protective — a high-pressure cutout guarding the blower, an O2 or LEL interlock, a motor overload — leave them alone. The legitimate move for a genuine nuisance trip is to retune one setpoint inside the equipment's rated limit, one change at a time, with the old and new values written down and a reason logged. Never defeat a gas-detection or overpressure interlock to make a number turn green. On a site making methane and H2S, that interlock is the thing standing between a stuck damper and a bad day.

Independent note: on aeration systems the ugly faults are the ones that get handed between the media contractor, the blower vendor, and whoever wrote the PLC code — each points at the other. An independent second look usually helps most right there, at the seam, where the pressure trend and the drive log and the control narrative finally get read side by side.